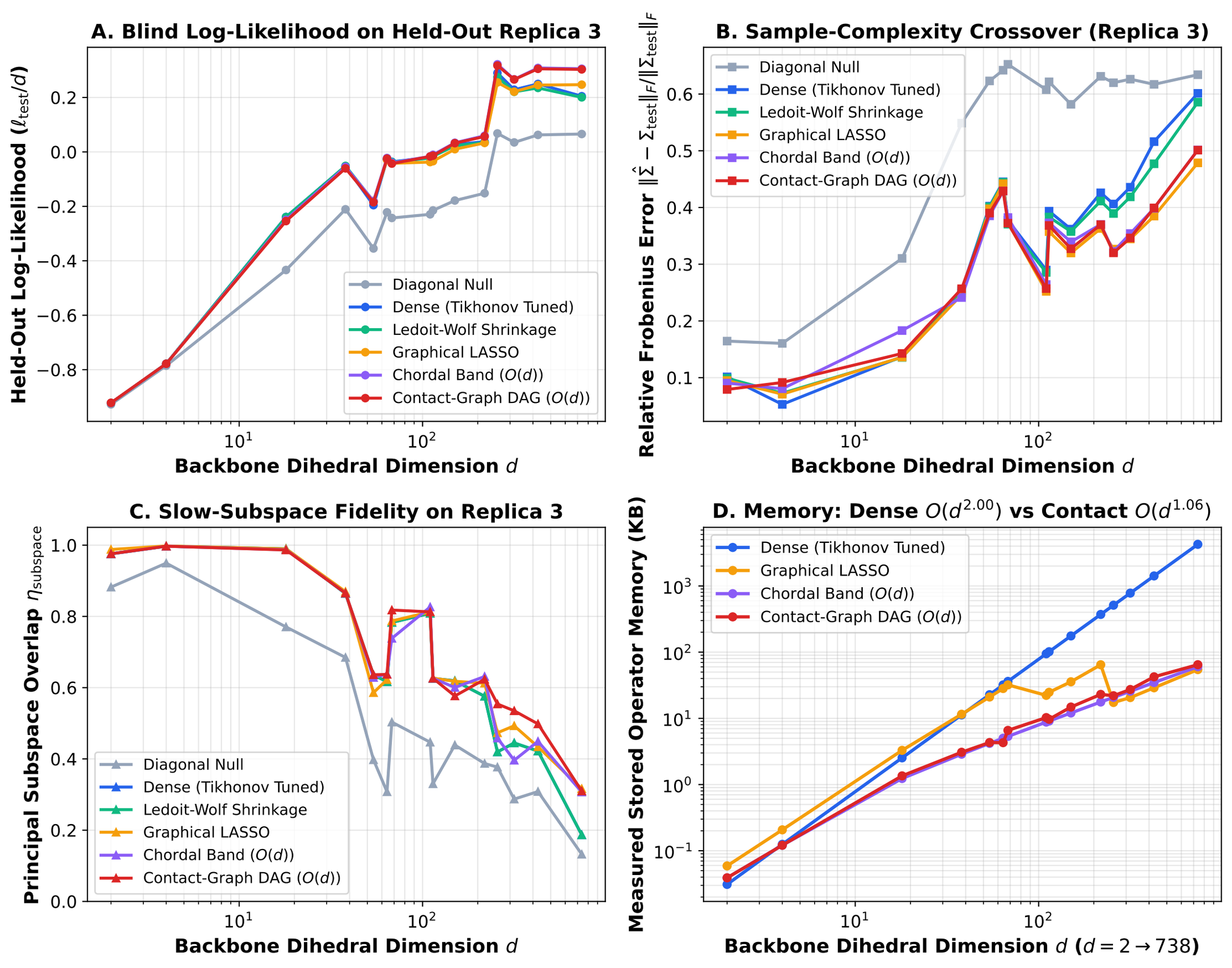

A. Blind Log-Likelihood on Held-Out Replica 3
Held-Out Log-Likelihood (ℓ_{test}/d)
Backbone Dihedral Dimension d
Diagonal Null
Dense (Tikhonov Tuned)
Ledoit-Wolf Shrinkage
Graphical LASSO
Chordal Band ($O(d)$)
Contact-Graph DAG ($O(d)$)
B. Sample-Complexity Crossover (Replica 3)
Relative Frobenius Error $\|\hat{\Sigma} - \Sigma_{\text{test}}\|_F / \|\Sigma_{\text{test}}\|_F$
Backbone Dihedral Dimension d
C. Slow-Subspace Fidelity on Replica 3
Principal Subspace Overlap η_{subspace}
Backbone Dihedral Dimension d
D. Memory: Dense $O(d^{2.00})$ vs Contact $O(d^{1.06})$
Measured Stored Operator Memory (KB)
Backbone Dihedral Dimension d ($d = 2 \to 738$)